

Ben Phan

 [phanben110.github.io](https://github.com/phanben110)  [phanben110](https://github.com/phanben110)  [benphan110](https://www.linkedin.com/in/benphan110)  Google Scholar
 phanben110@gmail.com  +(886) 984-305-278  Hsinchu City, Taiwan

EDUCATION

National Cheng Kung University (NCKU)

Sep 2023 – Aug 2025

Master of Science in Computer Science

- Advisor: [Prof. Jung-Hsien Chiang](#)
- Vice President of the Vietnamese Student Association at NCKU (2024 - 2025)
- Outstanding Thesis Award, invited to meet the Vice President at the Presidential Office of Taiwan

Danang University of Science & Technology

Sep 2018 – Mar 2023

Bachelor of Engineering in Control Engineering and Automation

- Advisor: [Ph.D. NGO Dinh-Thanh](#)
- Class President (2019 - 2023)
- Vice President of the BK Maker Scientific Research Club (2021 - 2022)

RESEARCH INTERESTS

My research interests include Artificial Intelligence, Machine Learning, and Deep Learning, with a focus on Natural Language Processing, Computer Vision, and AI Agents. I work on optimizing Large Language Models (LLMs) and deep learning architectures for efficient deployment on edge and local devices, exploring techniques such as model compression, retrieval-augmented generation, and knowledge graph integration to enhance reasoning, scalability, and performance under resource constraints.

PUBLICATIONS AND SUBMISSIONS

[MyMy at SemEval-2025 task 9: A robust knowledge-augmented data approach for reliable food hazard detection](#)

Ben Phan, Jung-Hsien Chiang

Proceedings of the 19th International Workshop on Semantic Evaluation, an ACL Workshop, 2025

Best System at SemEval-2025 Task 9, ACL Workshop, Vienna, Austria.

[CAS: enhancing implicit constrained data augmentation with semantic enrichment for biomedical relation extraction and beyond](#)

Fang-Yi Su, Gia-Han Ngo, Ben Phan, Jung-Hsien Chiang

Journal of Database (Q1), 2025

[Optimized biomedical entity relation extraction method with data augmentation and classification using GPT-4 and Gemini](#)

Cong-Phuoc Phan, Ben Phan, Jung-Hsien Chiang

Journal of Database (Q1), 2024

[Probability Model with Ensemble Learning and Data Augmentation for Named Entity Recognition \(NER\) and Relation Extraction \(RE\) Tasks](#)

Cong-Phuoc Phan, Gia-Han Ngo, Ben Phan, and Jung-Hsien Chiang

Proceedings of the BioCreative VIII: Curation and Evaluation in the era of Generative Models, 2023

Ranked among the Top 3 overall in the BioCreative VIII Challenge 2023.

Deep learning framework applied for predicting anomaly of respiratory sounds

Dat Ngo, Lam Pham, Anh Nguyen, **Ben Phan**, Khoa Tran, Truong Nguyen

International Symposium on Electrical and Electronics Engineering (ISEE), 2021

EXPERIENCE

Research Development Software Engineer

群聯電子 PHISON Electronics

Hsinchu, Taiwan

Sep 2025 –Now

- Develop firmware for SSD controller, focusing on Flash Physical Layer (FPL), error handling, and decoding flow.
- Research AI agents and LLM-based approaches for firmware development, aiming to improve automation, debugging, and system optimization.

AI Researcher

Intelligent Information Retrieval Lab (IIR Lab), CSIE, NCKU

Tainan, Taiwan

Sep 2023 –Aug 2025

- Developing and refining RAG techniques for enhanced information retrieval.
- Investigating Graph-based RAG models to improve the accuracy and efficiency of retrieval.
- Exploring integration of Large Language Models with retrieval systems to enhance the quality of generated information.

AI Engineer

Vingroup Big Data Institute

Hanoi, Vietnam

Jun 2022 –Sep 2023

- Completed courses in Linear Algebra, Probability & Statistics, Machine Learning, Deep Learning, Generative Model, Computer Vision, Natural Language Processing, and AI Ethics.
- Developed Face ID, Car Parking, and Uncovered Truck detection for VinFast factories and VinHome, processing data from 2,000+ cameras using GStreamer, DeepStream, TensorRT, and Triton.
- Optimized and deployed computer vision models (QAT, PQT) for VinFast Smart Factory, improving AI performance in industrial automation.

AI Engineer Internship

FTECH CO., LTD

Danang, Vietnam

May 2021 –Jun 2022

- Gained foundational knowledge of AI, deep learning, and computer vision techniques.
- Acquired hands-on experience with cloud platforms such as Azure and AWS for deploying AI models and services.

Embedded Software Internship

Da Nang Software Development Centre

Danang, Vietnam

May 2019 –Jun 2020

- Developed IoT-based tracking system for real-time bus movement monitoring in Danang.
- Gained proficiency in C, C++, and Linux environments through embedded software projects.

SKILLS

Computer Languages: Python, C++, C, Java, Bash, Linux.

Machine Learning Libraries: Pytorch, Tensorflow, Sklearn, Numpy, Triton, TensorRT, DeepStream, Gstreamer, Transformers, Langchain, LLMs, RAG.

Cloud & Tools: AWS & Azure, Docker, Git.

Languages: Vietnamese, English, Chinese.

ACHIEVEMENTS

🎓 Outstanding Thesis Award (Top 1%) 18th TSC AI Application Competition	2025
🎓 NCKU Distinguished International Student Scholarship for Master's program.	2023 – 2025
🏆 Best System at SemEval-2025 Task 9, ACL Workshop, Vienna, Austria.	2025
🏆 Ranked among the Top 3 overall in the BioCreative VIII Challenge.	2024
🎓 Academic Excellence Scholarship (Top 5%) at UD - DUT for outstanding performance	2023
🏆 Third Prize in USAID BUILD-IT & Dow Vietnam STEM.	2023
🎓 Vingroup AI Engineer Training Scholarship, selected for prestigious AI training program.	2022
🏆 Second Prize in Science Research, UD - DUT.	2022
🏆 Second Prize in BK-Techshow, UD - DUT.	2022
🏆 Top 8 in Siemens Vietnam SIMATIC Digital Transformation Contest.	2021
🏆 Appropriate Technology Award, University of Tsukuba, Japan.	2021
🏆 Top 6 & Creativity Award in ABU Robocon Asia-Pacific.	2020
🏆 Certificate of Merit in COVID-19 Community Service Project.	2020
🏆 Learning Express Program, Singapore Polytechnic Exchange.	2019

REFERENCES

Prof. Jung-Hsien Chiang

Distinguished Professor, Department of Computer Science and Information Engineering, NCKU.

Ph.D. NGO Dinh-Thanh

Lecturer at The University of Danang, University of Science and Technology.